

Misattribution Machine: AI-Driven Detection of Archival Misattribution

Technical Documentation — Exhibition System

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1. Project Overview

The Misattribution Machine is an exhibition system designed to detect and visualise the propagation of misattribution errors across digital archives and AI language models. The project analyses how incorrect nationality labels assigned to five artists — Kazimir Malevich, Alexandra Exter, Vera Pagava, David Kakabadze, and Sergei Parajanov — originate in archival authority records and are subsequently reproduced by major AI systems including Claude, GPT-4o, and LLAMA.

The system comprises three principal components:

- (1) A dataset analysis pipeline processing 659 archival records;
 - (2) A real-time AI misattribution detection tool using the Claude API; and
 - (3) A 12-screen LED exhibition installation rendered in p5.js.
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2. Dataset Analysis

Two datasets were processed to characterise the nature and scale of misattribution across institutional archival records.

2.1 Parajanov Bibliographic Dataset

305 records from bibliographic sources relating to Sergei Parajanov were analysed. The misattribution breakdown is as follows:

Misattribution Type	Record Count	Percentage
Wrong nationality label	218	71.5%
Omission of national identity	54	17.7%
Spelling / name variants	33	10.8%
Total	305	100%

2.2 Artworks Dataset (Malevich, Exter, Pagava, Kakabadze)

354 records across four visual artists were processed. The breakdown by artist and misattribution type is shown below:

Artist	Total Records	Wrong Label	Omission	Spelling / Structural
Kazimir Malevich	272	195	61	16
Alexandra Exter	44	38	4	2
Vera Pagava	19	19	—	—
David Kakabadze	19	—	—	19 (structural)
Subtotal	354	252	65	37

2.3 Combined Dataset Summary

Metric	Value
Total records analysed	659
Artists covered	5
Records with at least one form of misattribution	~94%

2.4 Artist Profiles and Misattribution Context

Kazimir Malevich. Born in Kyiv, 1879. Routinely labelled ‘Russian avant-garde’ or ‘Soviet artist’ in major archives including the Library of Congress, WorldCat, and MoMA, erasing his Ukrainian origin.

Alexandra Exter. Born in Kyiv; trained at the Kyiv School of Art. Labelled ‘Russian’ across most institutional records, with no reference to her Ukrainian formation.

Vera Pagava. Born in Tbilisi, Georgia. Labelled ‘Empire Russe’ in archival records — a framing that subsumes and erases Georgian identity entirely.

David Kakabadze. Born in Kutaisi/Tbilisi, Georgia. Absorbed into the Société Anonyme ‘Russian avant-garde’ canon without acknowledgement of Georgian nationality.

Sergei Parajanov. Soviet-Armenian filmmaker, Georgian-born. Birth name: Sarkis Hovsepi Parajanov. Labelled ‘Soviet’ in most records, with both national and ethnic identity omitted.

3. Propagation Chain

Misattribution errors propagate through a structured hierarchy of institutions. An absence of, or error within, a Library of Congress authority record is adopted as authoritative by downstream institutions, which in turn supply training data to AI language models. Those models subsequently reproduce and amplify the original error at scale.

Tier	Institution / System	Role in Propagation Chain
Tier 0	Library of Congress	Authority records — original source of misattribution
Tier 1	WorldCat / OCLC	Aggregation and global redistribution of authority labels
Tier 2	National libraries (Trove / NLA); Museum databases (MoMA, NGA, NGV)	Institutional adoption; misattribution embedded in catalogue data
Tier 3	AI language models (Claude, GPT-4o, LLaMA); Web knowledge graphs (Google Knowledge)	Trained on misattributed corpus; errors reproduced and amplified at scale

4. AI Detection System

The detection tool queries AI language models regarding an artist’s nationality and compares the model’s response against the Record of Truth derived from the dataset. Detected misattributions are flagged and, where possible, their provenance is traced to the originating authority record.

4.1 Technical Implementation

Component	Specification
Language	Python 3.9+
Primary API	Anthropic Claude API (claude-opus-4-5)
Multi-model	Claude (Anthropic), GPT-4o (OpenAI API), LLAMA (Ollama, local)
Detection method	Keyword matching against correct and incorrect nationality terms
Image queries	Vision queries supported via artwork image URLs

4.2 Server Routes

The HTTP server is implemented using the Python standard-library HTTP server (no external web frameworks). The following API routes are exposed:

Route	Method	Description
/api/query	GET	Single model query with misattribution assessment
/api/compare	GET	All three models compared side by side
/api/detect	GET	Misattribution detection for a given image URL
/api/artworks	GET	Full dataset records (filterable by artist)
/api/stats	GET	Dataset summary statistics
/api/images	GET	Image manifest and status

4.3 Source Files

File	Purpose
server/server.py	Main HTTP server
server/multi_llm.py	Multi-model comparison (Claude + GPT-4o + LLAMA)
server/claude_detector.py	Standalone CLI detection tool
server/image_pipeline.py	Artwork image management

5. Exhibition System

The exhibition is designed for a 12-screen LED installation configured as a 4-column × 3-row grid. Each screen is a separate browser window displaying one panel of a continuous full-canvas animation rendered in p5.js. Together the twelve panels form a single coherent visual field across the physical installation.

5.1 Visual Language

Each artwork is rendered as a cloud of 1,800 luminous particles, each positioned at a pixel sampled from the source image. Nearby particles are connected by fine threads. The visual language operates across two opposing states:

Truth state: Particles correctly positioned; threads form the image clearly; warm cream luminosity on a pure black background.

Misattribution state: Particles displaced by a noise force field; threads tangled; cold blue-white colouration indicating corrupted data.

Motion blur trail achieved via semi-transparent frame accumulation.

5.2 Phase Cycle (~35 seconds per artist)

Phase	Duration	Description
1. PRISTINE	3.5 s	Particles coalesce from void. Artwork emerges as constellation of light.
2. LABEL APPLY	3.5 s	Thin archival rectangle descends over the image.
3. CORRUPT	4.5 s	Noise force field tears particles from correct positions. Threads tangle.
4. PROPAGATE	5.0 s	Particle streams cross screen boundaries. Archive network activates.
5. MACHINE GAZE	4.5 s	Cold scanner light sweeps. AI systems read the corrupted data.
6. INTERVENTION	3.5 s	Warm light sweeps in. Spring force reclaims particles.
7. TRUTH	6.0 s	Particles lock to correct positions. Threads form the real image clearly.
8. FADE	2.0 s	Dissolves. Next artist begins.

5.3 Screen Synchronisation

Synchronisation across the twelve screens is achieved via the browser LocalStorage API. Screen 1 acts as the leader, writing phase state and timing data every 500 milliseconds. All other screens read from LocalStorage and align their animation phase accordingly. No server-side coordination infrastructure is required.

5.4 Screen URLs — 12-Screen Setup

Each screen is addressed via URL query parameters specifying its index within the 4 × 3 grid (screen number, column count, row count).

Screen	URL
Screen 1	http://localhost:8000/index.html?screen=1&cols=4&rows=3
Screen 2	http://localhost:8000/index.html?screen=2&cols=4&rows=3
Screen 3	http://localhost:8000/index.html?screen=3&cols=4&rows=3
Screen 4	http://localhost:8000/index.html?screen=4&cols=4&rows=3
Screen 5	http://localhost:8000/index.html?screen=5&cols=4&rows=3
Screen 6	http://localhost:8000/index.html?screen=6&cols=4&rows=3

Screen 7	<code>http://localhost:8000/index.html?screen=7&cols=4&rows=3</code>
Screen 8	<code>http://localhost:8000/index.html?screen=8&cols=4&rows=3</code>
Screen 9	<code>http://localhost:8000/index.html?screen=9&cols=4&rows=3</code>
Screen 10	<code>http://localhost:8000/index.html?screen=10&cols=4&rows=3</code>
Screen 11	<code>http://localhost:8000/index.html?screen=11&cols=4&rows=3</code>
Screen 12	<code>http://localhost:8000/index.html?screen=12&cols=4&rows=3</code>
Single-screen preview	<code>http://localhost:8000/index.html</code>
Curator controller	<code>http://localhost:8000/exhibition.html</code>

6. Visitor Kiosk

The visitor kiosk (`kiosk.html`) provides a touch-screen interface for gallery visitors. Upon selecting an artist, the system queries three AI models in parallel and presents their responses alongside the Record of Truth derived from the dataset.

URL: `http://localhost:8000/kiosk.html`

6.1 Interface Layout

The kiosk interface is divided into four columns: Claude | GPT-4o | LLAMA | Record of Truth. Each AI response is revealed via a typewriter text animation. A consensus bar indicates the degree of agreement or disagreement between the three models. Artist selector buttons allow navigation between Malevich, Exter, Pagava, Kakabadze, and Parajanov.

6.2 Operating Modes

Mode	Requirement	Behaviour
Demo mode	None (no API keys)	Pre-written responses demonstrate typical misattribution patterns from the dataset
Live mode	ANTHROPIC_API_KEY environment variable	Queries AI models in real time. GPT-4o and LLAMA also require their respective API keys

7. Technical Requirements

7.1 Dependencies

Requirement	Detail
Python	3.9 or higher
Python packages	anthropic, openai, openpyxl, gdown
Environment variable	ANTHROPIC_API_KEY (required for live AI detection)
Browser	Chrome or Firefox (modern version recommended)
Ollama	Required for LLAMA local inference (optional)

7.2 Installation and Start

Install Python dependencies:

```
pip install anthropic openai openpyxl gdown
```

Set environment variable and start the server:

```
export ANTHROPIC_API_KEY="your-key-here"
```

```
python3 server/server.py
```

7.3 Repository

Field	Value
Repository	github.com/yamanidzet/Teona-Yamanidze
Branch	claude/analyze-art-datasets-fvjZt

8. Visual References

The visual aesthetic of the Misattribution Machine draws from a lineage of computational art practices in which data is rendered as embodied, luminous matter rather than abstract information.

Memo Akten — The Networked Condition. Particle and thread systems in which human figures emerge from, and dissolve back into, data on a black ground. Akten's work establishes the visual grammar of luminous particles as carriers of identity.

Mario Klingemann. Painterly dissolution of recognisable images into constituent data points. Klingemann's aesthetic of controlled entropy informs the CORRUPT and PROPAGATE phases of the exhibition cycle.

Palette and Rendering

Element	Colour / Technique
Background	Pure black — the archive as void
Truth / clean state	Luminous warm-cream threads
Misattribution state	Cold blue-white threads
Motion blur	Semi-transparent frame accumulation (p5.js)
AI scanner sweep	Cold white light — machine reading
Intervention / correction	Warm amber sweep — human reclamation